

Project Title
(Mini Search Engine / Information-Retrieval System)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Mini Search Engine / Information-Retrieval System is a project focused on designing and implementing a compact yet efficient search engine capable of retrieving relevant information from a given collection of documents. The primary objective of the system is to demonstrate how real-world search engines organize, process, and retrieve information from large document collections. The system first builds an index for the document corpus, allowing user queries to be processed more efficiently than by repeatedly scanning every document. By combining suitable indexing, searching, ranking, and similarity techniques, the system aims to provide accurate and relevant search results while maintaining reasonable computational efficiency.

The proposed system supports four major classes of queries: exact keyword search, phrase/substring search, fuzzy search, and document-similarity search. Exact keyword search retrieves documents containing the specified terms, while phrase or substring search identifies documents containing a particular sequence or portion of text. Fuzzy search handles spelling mistakes and approximate matches by comparing the query with indexed terms and suggesting closely matching words. Document-similarity search identifies documents that are similar to a given document or query by representing documents using suitable feature vectors and calculating their similarity. The results obtained from these different query types are ranked according to their relevance, ensuring that the most useful documents are presented first. This combination allows the system to demonstrate both traditional keyword-based retrieval and more flexible similarity-based retrieval.

A major focus of the project is the manual implementation of the core algorithms involved in indexing, searching, ranking, fuzzy matching, and similarity computation rather than relying entirely on existing search-engine libraries. Appropriate data structures and algorithms are selected to improve retrieval performance and reduce unnecessary processing. The project also provides an analysis of the time and space complexity of each major algorithm, helping to justify the design decisions and understand how the system behaves as the size of the document corpus increases. Through this implementation, the project provides practical knowledge of information retrieval, inverted indexing, query processing, string matching, approximate matching, document ranking, and similarity measurement. Overall, the Mini Search Engine serves as a practical demonstration of how the fundamental components of a search engine can be designed and implemented from the ground up while balancing accuracy, efficiency, scalability, and algorithmic complexity.

Signature of the Guide